

Reasoning, Not Debate, Repairs Encoding-Fragile Graph Reasoning in LLMs

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Abstract

Large language models answer the same graph question with sharply different accuracy depending only on how the graph is serialized into text—a robustness gap we confirm at up to 0.47 on Llama-3.3-70B. A natural remedy is multi-agent *debate*: a Critic checks a Proposer’s reasoning before it is committed. Debate rests on a *verification asymmetry*—checking a claim should be cheaper than producing one—and graph reasoning is an unusually clean place to test it, because an edge claim is objectively present in the serialization or it is not. Across three tasks, three encodings, and three seeds (5,400 instances per condition), we find that debate does raise accuracy (+0.05) and does narrow the encoding gap. However, decomposing the debate into its reasoning scaffold and its verify-and-revise loop shows that the entire gain comes from asking the model to reason: the loop itself contributes −0.02. A compute-matched majority vote over independent reasoned answers beats debate on every cell at $1.07\times$ its token cost. The Critic genuinely detects errors ($\phi = +0.36$), but because correct answers outnumber wrong ones roughly five to one, its false alarms cost more accuracy than its detections repair. Encoding fragility is real and reasoning reduces it—but the multi-agent structure is not why.

1 Introduction

Large language models (LLMs) are increasingly asked to reason over graph-structured data—social networks, knowledge graphs, dependency structures—presented as text. A recurring and uncomfortable finding is that their accuracy is highly sensitive to *how* the graph is written down: the same graph and the same question, serialized under a different but equally faithful convention, can move accuracy by tens of points (Fatemi et al., 2024). We reproduce this *encoding fragility* on Llama-3.3-70B at a striking mag-

nitude: a `connected_nodes` query answered correctly 95.8% of the time under one serialization drops to 48.8% under another, on the same graphs. A model whose answer depends this much on a cosmetic choice of notation is not reasoning about the graph so much as about its surface form.

A natural remedy is to make the model check its own work. Multi-agent *debate*, in which a Proposer produces an answer and a Critic scrutinizes it before it is committed, has been reported to improve factuality and reasoning (Du et al., 2023; Irving et al., 2018). Debate rests on an implicit *verification asymmetry*: recognizing that a claim is wrong is assumed to be cheaper than producing the right claim in the first place. Yet the evidence is mixed—LLMs often cannot reliably self-correct (Huang et al., 2024), and when debate is placed head-to-head against simple aggregation it does not always win (Choi et al., 2025). A difficulty in adjudicating these results is that in most reasoning domains the intermediate steps a Critic must check are not *objectively* verifiable, so a failure of debate cannot be separated from a failure of checking.

Graph reasoning removes that obstacle. An atomic claim in a Proposer’s trace—“node 5 is adjacent to node 13”—either appears in the serialized graph or it does not, so a Critic’s job is objectively well-posed and its behavior can be measured exactly. This makes graphs an ideal testbed for the question we ask: **does debate reduce encoding fragility, and if it helps at all, is the multi-agent structure the reason?**

We run three graph tasks (`edge_existence`, `node_degree`, `connected_nodes`) under three encodings (`adjacency`, `incident`, `friendship`) across three independent seeds, comparing a zero-shot baseline, a Proposer–Critic debate, and a compute-matched majority vote. Our findings are:

- **Encoding fragility is large and significant** on all three tasks, unanimously worst under `friendship` and best under `incident` (§4).

- **Debate raises accuracy and narrows the encoding gap**, replicating across all three seeds—the result one would report by stopping here.
- **The gain is the reasoning scaffold, not the debate.** Decomposing debate into its first-turn chain-of-thought and its verify-and-revise loop, the loop contributes -0.02 : asking the model to reason helps, having a Critic argue about the reasoning does not.
- **A compute-matched majority vote beats debate** on every cell at $1.07\times$ its token cost. The Critic does carry real signal, but its revisions are net negative because of a precision, not a detection, problem.

The upshot is a positive result wrapped around a negative one: encoding fragility is real and reasoning genuinely reduces it, but the multi-agent debate structure—the part that is expensive and elaborate—is not what does the work.

2 Related Work

LLM reasoning over serialized graphs. A body of recent work studies how well LLMs answer questions about graphs given as text, and consistently finds that the choice of *encoding*—how nodes, edges, and structure are rendered into a string—is a first-order determinant of accuracy. [Fatemi et al. \(2024\)](#) systematically vary graph encodings and show that model performance swings widely across them, with no single encoding dominant across tasks; [Perozzi et al. \(2024\)](#) pursue the complementary direction of learning structured encodings rather than fixing a textual one. This *encoding fragility* is the phenomenon our work takes as its starting point: we treat cross-encoding accuracy spread as a robustness metric and ask whether a reasoning procedure can shrink it.

Multi-agent debate. Debate frames inference as an interaction between models that argue toward a better answer. [Irving et al. \(2018\)](#) propose debate as a mechanism for supervising systems whose reasoning is hard to check directly, and [Du et al. \(2023\)](#) report that having multiple LLM instances propose and critique across rounds improves factuality and commonsense and arithmetic reasoning. Our Proposer–Critic setup is a deliberately minimal, two-role instance of this family: a single model produces one answer, then critiques and revises it. Mechanically this is a form of *self-correction*, whose reliability is itself contested—[Huang et al. \(2024\)](#) find that LLMs largely cannot self-correct

reasoning without an external signal. Crucially, that prior work cannot say whether the failure is one of *checking* or of *acting on* a correct check, because its intermediate claims are not objectively verifiable. The verification asymmetry underlying debate predicts that an objectively checkable domain is exactly where the loop should finally help; our graph setting makes the claims checkable, so we can run that test and measure the Critic’s detection and false-alarm rates directly rather than only observing an outcome.

Aggregation as the alternative. A simpler way to spend extra inference compute is to sample several answers independently and aggregate them. Self-consistency ([Wang et al., 2023](#)) marginalizes over multiple chain-of-thought ([Wei et al., 2022](#)) samples by majority vote and reliably improves reasoning accuracy. This makes voting the natural control for debate: both spend more compute than a single answer, but debate lets the attempts *interact* while voting keeps them independent. [Choi et al. \(2025\)](#) compare the two and find voting competitive with or better than debate in several settings. We sharpen this comparison in two ways: we *match compute* between the two arms by measured token cost rather than by response count, and we run it on a task where fragility gives debate its best chance—an objectively checkable domain in which the Critic provably discriminates right from wrong.

3 Setup

Data. We generate undirected Erdős–Rényi graphs with NetworkX ([Hagberg et al., 2008](#)), 5 to 19 nodes each and edge probability drawn per graph, giving a wide spread of densities. For each of three seeds (7, 11, 13) we draw an independent set of 200 graphs, and every graph is rendered under all three encodings, so a per-encoding accuracy difference reflects the encoding alone and all comparisons across encodings are *paired* by graph.

Tasks and encodings. Three tasks span three answer types: `edge_existence` (is there an edge between two given nodes? — boolean), `node_degree` (how many neighbors does a node have? — integer), and `connected_nodes` (which nodes is a given node adjacent to? — a set). Each graph is serialized under three encodings from [Fatemi et al. \(2024\)](#): adjacency, an edge list; incident, per-node neighbor lists; and friendship, natural-language statements. The

Graph	G :	nodes	0–6;	edges
		(0, 1)(0, 2)(2, 4)(2, 6)(4, 5)(5, 6); node 3 isolated.		
adjacency		The edges in G are: (0, 1) (0, 2) (2, 4) (2, 6) (4, 5) (5, 6).		
incident		Node 0 is connected to nodes 1, 2. Node 2 to 0, 4, 6. Node 4 to 2, 5. Node 5 to 4, 6. Node 6 to 2, 5.		
friendship		James and Robert are friends. James and John are friends. John and David are friends. John and Patricia are friends. David and Mary are friends. Mary and Patricia are friends.		
Tasks (query shown under adjacency; gold answer in <i>italics</i>):				
edge_existence		Is node 6 connected to node 1? <i>No.</i>		
node_degree		What is the degree of node 5? <i>2.</i>		
connected_nodes		List all nodes connected to 2. <i>0, 4, 6.</i>		

Table 1: One graph under the three encodings, and the three tasks. The encodings are information-preserving re-serializations of the same graph; friendship maps nodes 0–6 to James, Robert, John, Michael, David, Mary, Patricia. A shared preamble naming the nodes precedes each encoding and is omitted here.

first two label nodes with integers, friendship with popular names; Table 1 shows one graph under all three, together with the three tasks. The encodings preserve the graph exactly and differ only in surface form. Crossing 3 tasks $\times$ 3 encodings $\times$ 3 seeds $\times$ 200 graphs gives 9 cells of $n = 600$ each, 5,400 instances per condition.

Model and conditions. All conditions use Llama-3.3-70B-Instruct-Turbo (FP8) served by Together.ai. We compare three (Table 2). The *baseline* is a single zero-shot answer-only response. *Debate* is a Proposer–Critic loop: the Proposer emits a numbered claim trace and a final answer, the Critic reviews it against the encoding and returns AGREE or REVISE, and the Proposer revises until agreement or a turn budget is reached; the Proposer and Critic are the *same* model in different roles. Its first turn is a single-turn chain-of-thought answer under the same decoding, which lets us separate the reasoning *scaffold* (baseline $\rightarrow$ turn 1) from the verify-and-revise *loop* (turn 1 $\rightarrow$ final) with no extra runs. *Majority vote* draws the same Proposer prompt $N=3$ times independently at the model’s shipped sampling defaults and takes the mode of the parsed answers; it differs from debate in exactly one thing—whether the attempts see each other.

Condition	Decoding	Cap	Cost/inst.
Baseline	greedy	256	1 resp / 364 tok
Debate	greedy	512	2.3 resp / 1,554 tok
Majority vote	$T=0.6$	512	3 resp / 1,664 tok

Table 2: The three conditions. Compute is matched by measured token cost, not by response count: the vote spends $1.07\times$ debate’s tokens.

Matched compute. $N=3$ was fixed *before* the run by matching debate’s measured token spend (debate buys 2.6–2.9 turn-1-equivalents per instance). The realized cost is 1,664 tokens per instance for the vote against 1,554 for debate, a ratio of $1.07\times$, so the vote is if anything the cheaper arm and any advantage it shows is not bought with extra compute. The sampling temperature is Llama’s shipped default, chosen before seeing any result; realized answer diversity is reported as a diagnostic, never tuned.

Metrics and statistics. The primary metric is accuracy; the secondary is *encoding fragility*, the cross-encoding accuracy spread (max – min) within a task. Because encodings are paired by graph, significance uses paired tests: Cochran’s Q as an omnibus across a task’s three encodings, and McNemar’s test for pairwise gaps and for condition-vs-condition accuracy differences; cell accuracies carry Wilson intervals. The Critic diagnostic (does a REVISE track the answer actually being wrong?) is unpaired, one observation per verdict, so it takes a χ^2 association test reported with the signed ϕ . We report the fragility spread descriptively, supported by replication across the three independent seeds rather than by a test of the spread itself. Finally, edge_existence is a single-pair lookup whose trace reduces to one atomic claim, so a claim-by-claim critique has nothing to work on and its baseline sits near ceiling; it behaves as a *control* rather than a third fragile task, and we report every result both over all 9 cells and over the 6 cells that exclude it, never dropping it silently.

4 Results and Analysis

4.1 Encoding fragility is large

Baseline accuracy swings sharply with the serialization on all three tasks. On connected_nodes it runs from 0.958 under incident to 0.488 under friendship—a 0.470 gap on the same graphs; node_degree spans 0.237 and even the near-ceiling edge_existence control spans 0.113. The

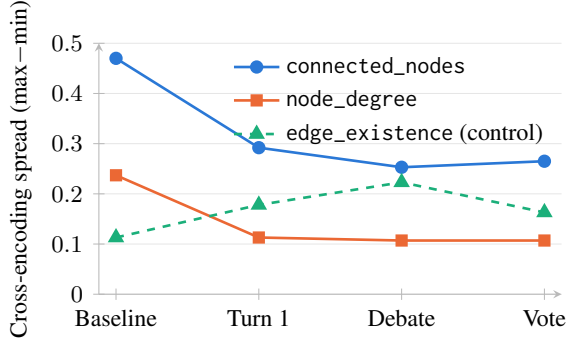


Figure 1: Cross-encoding accuracy spread (fragility) by stage; lower is less fragile. Reasoning (baseline → turn 1) collapses the spread on the two fragile tasks, and the verify-and-revise loop and the vote add little beyond it. On the edge_existence control the spread instead *rises*, as the loop degrades a near-ceiling task.

effect is highly significant under the paired omnibus (Cochran’s $Q = 377, 148, 81$ for the three tasks; all $p < 10^{-17}$), and its shape is unanimous: incident is the best encoding and friendship the worst on every task. friendship is the one encoding that names nodes rather than numbering them, which is consistent with named entities making a spurious relation easier to assert. This fragility is the phenomenon the rest of the paper asks a reasoning procedure to repair.

4.2 Debate helps—if you stop here

Read at the top line, debate looks like a success. It raises mean accuracy from 0.793 to 0.843 over all nine cells (+0.050; McNemar $b/c = 351/619$, $p = 10^{-17}$) and from 0.726 to 0.843 over the six cells excluding the edge_existence control (+0.117; $p = 10^{-54}$), and the gain replicates in sign across all three independent seeds. It also narrows the encoding gap (Figure 1): the connected_nodes spread falls $0.470 \rightarrow 0.253$ and node_degree $0.237 \rightarrow 0.107$, with the narrowing reproducing 3/3 on both fragile tasks. The gains concentrate exactly where accuracy was worst—the three largest per-cell gains are the three lowest baseline cells—so debate works by lifting the floor, which is what narrows the spread. A paper that reported only this would conclude that debate improves encoding-fragile graph reasoning.

4.3 The gain is the reasoning scaffold, not the loop

Debate’s first turn is a single-turn chain-of-thought answer under identical decoding, so the split baseline → turn 1 → final separates the reasoning *scaffold* from the verify-and-revise *loop* (Table 3). The scaffold accounts for the entire improvement: it is worth +0.071 over nine cells and +0.120 over six. The loop then gives back -0.021 and -0.003 . Excluding the edge_existence control does not rescue the loop; it moves it from *harmful* to merely *inert*. The loop helps in just one cell of nine and hurts in two. Both losses are on the edge_existence control, where the Critic drags accuracy from 0.922 to 0.803. That task has a single atomic claim: there is nothing to critique, so the Critic only talks a correct answer down. The one-line reading is that *asking the model to reason helps; having a Critic argue about the reasoning does not*.

4.4 A compute-matched vote beats debate

If the loop is not paying for its compute, the natural comparison is against spending that compute on independent attempts instead. Majority vote scores 0.875 over nine cells against debate’s 0.843 (+0.032; $p = 2 \times 10^{-16}$) and 0.853 against 0.843 over six (+0.010; $p = 0.035$), at $1.07\times$ debate’s tokens, and it *loses no cell*—it is at least as accurate as debate in all nine and significantly better in three. It beats debate in all three seeds. The full ordering across the arm is baseline $0.793 < \text{debate } 0.843 < \text{turn 1 } 0.864 < \text{vote } 0.875$, so debate is the *worst* of the three reasoning conditions and is beaten even by its own single first turn. The vote’s advantage is largest on edge_existence (+0.147 on adjacency), exactly where debate is destructive: an independent redraw cannot be argued out of a correct answer, whereas a Critic can. Because debate and the vote differ in only one thing—whether the extra attempts interact—this isolates *interaction* as the ingredient that is not paying off.

4.5 Why the loop fails: precision, not detection

The loop’s failure is not that the Critic cannot tell right from wrong. Cross-tabbing every verdict against whether the answer it judged was actually correct, a REVISE is strongly associated with a wrong answer (χ^2 on 5,825 verdicts, $\phi = +0.358$; a REVISE moves the probability the answer is wrong from the base rate 0.181 to 0.459), and the

Condition	All 9 cells	6 (no edge_ex.)
Baseline	0.793	0.726
Turn 1 (CoT)	0.864	0.846
Debate (final)	0.843	0.843
Majority vote ($N=3$)	0.875	0.853
Scaffold (base→T1)	+0.071	+0.120
Loop (T1→final)	−0.021	−0.003
Vote − debate	+0.032	+0.010

Table 3: Mean accuracy by condition, over all nine cells and the six excluding the edge_existence control. The scaffold (turn 1) carries the entire gain over the baseline; the verify-and-revise loop is negative-to-inert; the compute-matched vote is best on both scopes.

discrimination is strongest ($\phi = +0.69$) precisely on edge_existence, where the check is a single lookup. The Critic detects 50.2% of wrong answers. The problem is what it costs: it also raises a false alarm on 13.2% of *correct* answers, and because correct answers outnumber wrong ones roughly five to one, those false alarms convert into more damage than the detections repair. Net across the arm, revisions move 113 more answers from right to wrong than the reverse, and are net-positive in only one cell of nine. The loop fails on **precision**, not detection—the Critic knows the answer is wrong on edge_existence and still, by revising too many correct answers, makes that task worse.

5 Discussion

Reasoning repairs fragility; structure does not.

The encoding gap does shrink under debate, but the decomposition locates the cause: the single-turn reasoning scaffold produces essentially all of the narrowing, and a compute-matched vote reproduces it (Table 3). What reduces a model’s sensitivity to how a graph is written down is being asked to reason about it, not the multi-agent apparatus around that reasoning. We report the spread reduction descriptively—it replicates across all three seeds, but we do not test the spread itself—so the claim is that reasoning narrows the gap, not that debate confers encoding-invariance: fragility stays highly significant under every condition.

The asymmetry can hold and debate still lose.

Debate’s premise—that recognizing a wrong claim is cheaper than producing a right one—was satisfied here: the Critic discriminates right from wrong ($\phi = +0.36$, and $+0.69$ where the check is a single lookup). Yet the loop still lost accuracy, for a reason that is general. A verify-and-

revise step changes accuracy by its detection rate times the fraction of answers that are wrong, minus its false-alarm rate times the fraction that are right. When a model is already mostly correct—as Llama-3.3-70B is here, roughly five right for every one wrong—this balance tips negative unless the Critic’s precision is very high, however well it detects. Objective checkability, the ingredient prior self-correction studies lacked, was not enough: it sharpened detection, which was never the bottleneck, and left precision untouched. The lesson transfers beyond graphs: self-verification pays off only where errors are common or the check is near-perfectly precise, and neither holds for a strong model on a task it mostly solves.

Spend compute on independence, not interaction.

Because debate and the vote differ only in whether the extra attempts interact, the comparison is a clean test of the two ways to spend inference compute, and independence wins: three independent draws cannot argue each other out of a correct answer, whereas a Critic can. Given a fixed budget and a task the model already handles reasonably, sampling and voting is the safer use of it than a self-critique loop. This does not refute debate in general—it may still help where the base error rate is high, or where an external, reliable verifier replaces the model’s own Critic—but on encoding-fragile graph reasoning, at matched cost, it is dominated.

6 Limitations and future work

- **Single model and graph regime.** Our results come from one model (Llama-3.3-70B) on small Erdős–Rényi graphs. The net-effect account of §5 predicts the loop’s sign depends on the base error rate, so harder tasks and weaker models are the test of whether the failure generalizes.
- **Single prompt.** We use one Proposer/Critic wording, and the Critic’s precision may depend heavily on it; varying the prompts would separate a property of the method from one of the formulation.
- **Debate structure.** We tested one minimal two-role loop; richer structures—more agents, independent proposers that then interact, or multiple rounds—may recover value over voting.

7 Conclusion

LLM graph reasoning is sharply encoding-fragile, and asking the model to reason reduces that

fragility. A Proposer–Critic debate appears to help, but the gain is entirely the reasoning it elicits: the verify-and-revise loop is inert-to-harmful, and a compute-matched majority vote beats debate on every cell. The Critic genuinely detects errors; it loses because, against a mostly-correct model, its false alarms cost more than its detections repair. On this task, reasoning helps—and debate is not why.

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